

# MSFE-AD: Multiscale Feature-Enhanced Early Satellite Sensor Data Anomaly Detection Based on Rare Boundary Samples

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**Abstract**—With the increasing number of satellites being launched, in-orbit health monitoring has garnered significant attention. Analyzing operational data collected by spaceborne sensors enables the early detection of anomalies, thereby helping to prevent critical failures. However, due to the high morphological similarity between early fault data and normal data, achieving high detection rates while maintaining low false alarm rates remains a challenge. To address this issue, we propose a novel anomaly detection method that leverages rare boundary samples to guide key feature selection for feature augmentation, thereby constructing a highly discriminative feature space. The deep model incorporates a multifeature anomaly detection strategy to accurately identify abnormal data. By jointly extracting features from multiple domains and employing an adaptive weight allocation mechanism, the model enhances generalization across diverse data distributions. Experiments on real satellite sensor datasets demonstrate excellent detection performance, achieving a recall of 0.826, a precision of 0.938, and an *F1*-score of 0.878, indicating strong detection capability and robustness. Ablation studies confirm the effectiveness of integrating multiple feature augmentation strategies in balancing detection sensitivity and precision. These results highlight the proposed method's superior ability to detect early anomalies in satellite systems.

**Index Terms**—Anomaly detection, early anomalies, feature enhancement, satellite telemetry, sensor data.

## I. INTRODUCTION

ARTIFICIAL satellites are widely used in navigation, communications, weather forecasting, resource exploration, and other fields. Their long-term stable and reliable operation is crucial. Satellite sensor data, commonly referred to as telemetry data, are collected by spaceborne sensors to monitor satellite health. A large number of sensors are used to measure the working conditions of various engineering subsystems, such as satellite attitude, internal temperature,

pressure, voltage, and current. The collected data are transmitted to ground stations via radio technology for analysis by researchers. As the sole source of information for ground personnel to assess satellite status, these data must be accurately interpreted to ensure that operators can promptly and precisely understand the satellites operational condition [1].

During prolonged operation in space, satellite components may degrade and eventually fail due to aging. Before an actual malfunction occurs, sensor data often exhibit early signs of failure. Anomaly detection aims to identify these precursor signals, enabling engineers to take timely intervention measures and effectively prevent major spacecraft failures. Data-driven anomaly detection methods are now widely adopted. Researchers have employed a variety of machine learning and deep learning algorithms to detect anomalies in satellite sensor data [2]. Failures caused by gradual equipment degradation typically follow a progressive process, during which the severity of anomalies in sensor data increases incrementally. The earlier an anomaly is detected, the more beneficial it is for restoring normal satellite operation and extending its service life. Therefore, studying anomaly detection in the early stages of satellite sensor data holds significant practical value.

In the industrial field, scholars have conducted extensive studies on early anomaly detection in sensor data. Early anomaly and fault detection have become critical for ensuring operational safety and reducing maintenance costs [3]. Since early faults manifest as weak signal characteristics and high noise levels, numerous studies have focused on improving weak fault detection by suppressing broadband noise and periodic components in the signals [4], [5]. Traditional vibration analysis and statistical process control methods often fail to capture early faults [6]. Studies in [7] and [8] improve the signal-to-noise ratio through feature extraction methods, combined with deep learning models to detect anomalies. These methods offer high interpretability and computational efficiency; however, their ability to handle complex nonlinear features is limited.

Several recent studies have attempted to address these challenges by designing frameworks with high detection sensitivity [9]. For example, [10] proposed an unsupervised deep tensor framework with adaptive rule updates, which does not require labeled data and can adapt to device degradation in real time. Similarly, [11] incorporates an adversarial learning approach

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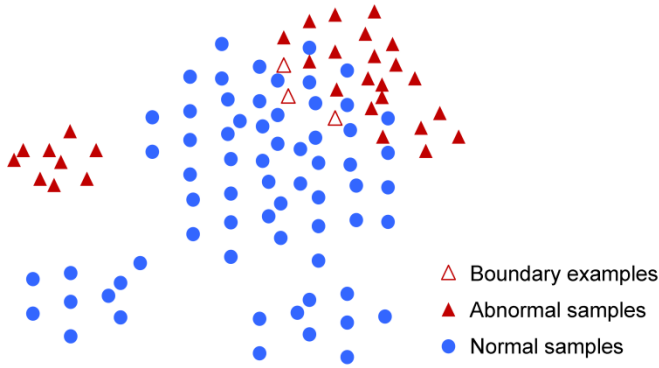


Fig. 1. Illustration of early anomaly data distribution.

with an autonomous model update mechanism to dynamically adjust the detection threshold. However, these approaches still have limitations in dealing with unbalanced data. Parallel convolutional neural network (CNN) architectures [12] enhance multimodal data processing capabilities, albeit with high computational overhead, while LSTM-based predictors [13] struggle to address error accumulation in long-term prediction. Such methods may improve anomaly detection but can also increase the false alarm rate. In addition, generalization ability may remain a challenge.

Early anomaly detection is challenging due to the high similarity between anomalous sensor data and normal data. Some anomalous data overlap with normal data and are difficult to distinguish (see Fig. 1); when there are no representative boundary samples, the model cannot learn to distinguish between normal and anomalous classes, and the samples near the decision boundary can be misclassified. In spacecraft anomaly detection systems, boundary information can lead to overly conservative thresholds or overly sensitive thresholds.

It is worth noting that during satellite operation and maintenance, engineers typically perform regular manual inspections of sensor data. Boundary samples are often manually identified by experts and are indicative of weak faults. However, these samples are often discarded due to the difficulty of incorporating them into anomaly detection models. Due to the diversity of anomaly patterns, anomaly detection is often approached as a semisupervised learning task. This approach enables the model to detect a wide range of anomalies, but it neglects boundary samples that contain valuable prior knowledge. In our ongoing observations of in-orbit satellite sensor data, we have found that some anomalous data missed by the detector exhibit similarities to boundary samples previously labeled by experts. Therefore, we propose to leverage these previously discarded boundary samples to enhance the performance of the anomaly detector. These samples are distributed near the decision boundary between the normal and anomalous classes in the feature space, but they occur infrequently and are commonly referred to as rare boundary samples.

Boundary samples play an important role in classification tasks, as they may influence the positioning of decision boundaries [14]. In particular, boundary oversampling techniques can enhance overall performance in imbalanced classification tasks [15]. These methods employ data augmentation techniques to

mitigate category imbalance and enhance the classifiers ability to recognize boundary samples [16]. However, several challenges remain with existing methods. The generated samples may be overly concentrated in certain regions, causing the model to rely excessively on localized features and resulting in overfitting. In addition, these methods are ineffective in handling the classification of multidimensional, multiclass, imbalanced data. Moreover, data augmentation techniques designed for time-series classification may not be directly applicable to time-series anomaly detection tasks [17].

To address the aforementioned challenges, we propose the MSFE-AD model, which fully leverages the prior knowledge contained in expert-labeled boundary samples during training. The proposed method effectively captures subtle anomalies induced by incipient faults.

The main contributions of this article are summarized as follows.

- 1) This article presents the first attempt to improve anomaly detection performance using rare boundary samples. A feature engineering method is introduced to significantly enhance data characterization by mining potential features from original data and selecting the most discriminative subset.
- 2) A multiscale anomaly detection strategy is proposed, employing parallel processing of heterogeneous detection models to improve detection precision and generalization ability. A learnable influence factor is introduced to adaptively adjust the contribution weights of different detection strategies, maintaining model stability in complex environments. This approach effectively balances the tradeoff between detection sensitivity and robustness, providing an interpretable and adaptable solution for early satellite anomaly detection.

The remainder of this article is organized as follows. Section II presents the methods of model reconstruction and feature enhancement for time-series anomaly detection. Section III describes the proposed framework and its implementation. Section IV presents experiments conducted on satellite sensor data, where the proposed method is compared with existing approaches. Section V discusses the experimental results and provides further analysis of the framework. Section VI concludes this article.

## II. IRELEVANT METHODOLOGIES

Sensor data are a form of time-series data. Time-series anomaly detection methods are generally categorized into prediction- and reconstruction-based approaches [18]. Prediction-based methods are sensitive to short-term fluctuations and are well-suited for real-time anomaly detection. In contrast, reconstruction-based methods are more effective in capturing trend deviations in time-series data. In this work, we adopt a reconstruction-based anomaly detection approach. Among various reconstruction models, autoencoders (AEs) and their variants have been widely adopted for this task.

### A. Anomaly Detection Methods Based on Reconstruction

AEs are a class of neural networks designed for data compression and reconstruction tasks. They are designed to

replicate input data through learned internal representations. An AE typically consists of an encoder and a decoder. The encoder maps the input into a latent representation, commonly referred to as the encoding, while the decoder reconstructs the input data from this latent space, referred to as the decoding. When trained on normal data, AEs typically exhibit large reconstruction errors when presented with anomalous inputs, making them well-suited for time-series anomaly detection.

For example, Kim et al. [19] proposed a Bayesian auto-optimization approach for AE hyperparameters, which enhances anomaly detection performance for boiler tube leakages by optimizing latent dimensionality. Rani et al. [20] introduced a probabilistic wavelet neural operator AE (PWNOAE) for anomaly detection in industrial process data. In this model, wavelets were employed for time-frequency localization of time-series data, while deep models were used to learn complex features from the data.

Variational AEs (VAEs) are generative models that combine the strengths of AEs and variational inference. Compared with traditional AEs, VAEs offer enhanced generative capabilities, more flexible latent representations, and improved probabilistic modeling. Li et al. [21] incorporated recurrent neural network (RNN) and gated recurrent unit (GRU) structures to model temporal dependencies and uncertainty in time series and employed a VAE to compute reconstruction loss, followed by a density-based detector for anomaly identification. To address the nonlinearity, dynamics, and non-Gaussian characteristics of industrial process data, Zhang et al. [22] proposed a recurrent Kalman VAE (RK-VAE), which integrates Kalman filtering for noise suppression.

Long short-term memory (LSTM) networks possess powerful nonlinear modeling capabilities and effectively capture temporal dependencies in sequential data. When combined with AEs or VAEs, the resulting models leverage the strengths of both structures, thereby improving their suitability for time-series anomaly detection. Han et al. [23] introduced an LSTM-based VAE (LSTM-VAE) with reconstruction probability for fault detection in ship components. Guha et al. [24] proposed an anomaly detection mechanism that first applies wavelet decomposition to extract high- and mid-frequency signals, thereby mitigating the impact of rare anomalies in the training data, and then uses a VAE-LSTM model for reconstruction-based detection in power grid data.

### B. Feature Enhancement Methods

In time-series analysis, feature enhancement can significantly improve model performance by emphasizing salient characteristics of the original data. Frequency-domain enhancement techniques improve the models capacity to capture periodicity and trends by extracting spectral features from time-series data. Frequency-enhanced channel attention mechanisms [25], which apply frequency-domain transformations to improve feature extraction in time-series prediction, emphasize the importance of multiscale frequency information. Dual-frequency augmented attention networks [26], which combine high- and low-frequency components to predict the remaining useful life of aircraft engines, demonstrate superior performance by capturing distinct frequency decay modes.

Self-supervised learning leverages the data itself to automatically generate labels, thereby improving the models ability to discriminate time-series features. TimesURL [27] introduces a self-supervised contrastive learning framework to improve both the quantity and quality of negative samples. Hu et al. [28] proposed a generalized representation model based on contrastive learning, employing recurrent generative adversarial networks (R-GANs) to generate positive and negative samples.

The multiscale fusion technique enhances model learning for complex time series by integrating features across different scales. Zhang et al. [29] introduced a hierarchical transformer structure to capture frequency-domain features at multiple time scales, enhancing multivariate time-series prediction through multiscale feature fusion. Zhong et al. [30] employed the discrete wavelet transform (DWT) to obtain multiscale time-series features, combined with an energy-based multiscale weighting strategy to improve the models transferability. These methods emphasize the importance of multiscale modeling in addressing complex time-series analysis tasks.

These research efforts provide a solid theoretical foundation for this work. The following section elaborates on the novel feature enhancement framework proposed herein.

## III. PROPOSED METHOD

In this section, the framework of MSFE-AD is introduced, including three components: feature engineering, reconstruction modeling, and detection strategy, as illustrated in Fig. 2. In the feature engineering module, both global and local features are extracted, and the most representative features are selected and reinforced in the data. The reconstruction model is implemented using a VAE, with LSTM networks employed for both the encoder and the decoder to reconstruct the input time series. The detection strategy employs two complementary methods at different time scales to enhance anomaly detection performance. An influence factor is introduced to adaptively weight the contribution of each detection strategy, maintaining a balance between them.

### A. Feature Engineering

Early satellite anomalies are challenging to detect due to their subtle and inconspicuous characteristics. In this article, we propose a feature enhancement method for time-series data, which enhances raw signals to amplify subtle anomalies, making them more distinguishable to the detection model.

1) *Feature Extraction*: Statistical, spectral-domain, and time-domain features are extracted to characterize the time series. These representative features effectively capture the underlying characteristics of the time series and possess clear physical interpretability. A detailed description of these three types of features is provided in Table I. These features are extracted at two different time scales: global features, which are computed over the entire time series and represented as single numerical values; and local features, which are obtained using a sliding window, resulting in a sequence of feature vectors.

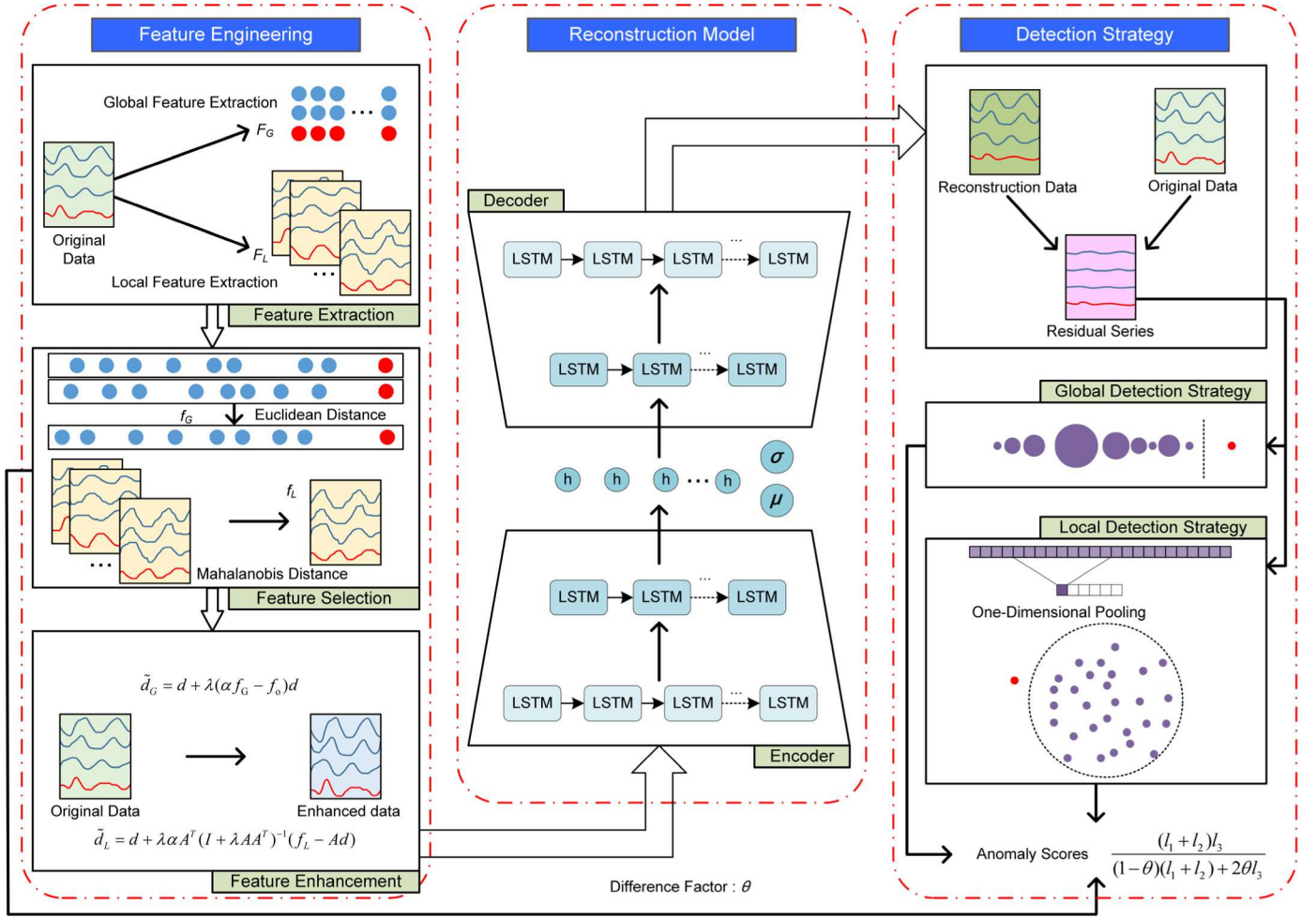


Fig. 2. MSFE-AD algorithmic framework.

TABLE I  
DETAILED INFORMATION ON THE THREE TYPES OF FEATURES

| Statistical Features    | Spectral-Domain Features   | Time-Domain Features        |
|-------------------------|----------------------------|-----------------------------|
| Mean                    | FFT Mean Coefficient       | Mean Differences            |
| Skewness                | Wavelet Absolute Mean      | Mean Absolute Differences   |
| Kurtosis                | Wavelet Standard Deviation | Median Differences          |
| Interquartile Range     | Wavelet Variance           | Median Absolute Differences |
| Mean Absolute Deviation | Spectral Distance          | Sum of Absolute Differences |
| Standard Deviation      | Spectral Median Frequency  | Entropy                     |
| ECDF Slope              | Spectral Maximum Peaks     | Peak to Peak Distance       |

2) *Feature Selection*: In the preceding step, global and local features were extracted from the time-series data. The extracted global features are represented as a set of discrete values.

Let  $F_G$  denote the collection of global features for all data, where  $k \in [1, 21]$  indicates the type of feature and  $n$  represents

the index of the data sample

$$F_G = \begin{Bmatrix} f_{1,1}, f_{1,2}, \dots, f_{1,n} \\ f_{2,1}, f_{2,2}, \dots, f_{2,n} \\ \vdots \\ f_{k,1}, f_{k,2}, \dots, f_{k,n} \end{Bmatrix} \quad (1)$$

where  $F_L$  represents the set of local features for all data and  $f'_{k,n}$  are features in the form of time series

$$F_L = \begin{Bmatrix} f'_{1,1}, f'_{1,2}, \dots, f'_{1,n} \\ f'_{2,1}, f'_{2,2}, \dots, f'_{2,n} \\ \vdots \\ f'_{k,1}, f'_{k,2}, \dots, f'_{k,n} \end{Bmatrix}. \quad (2)$$

Subsequently, the discriminative value of these features needs to be evaluated. At this stage, boundary samples are taken into account to guide feature selection. Typical representative features are identified based on their ability to separate anomalous data from normal data. The Euclidean distance is used for evaluation, and the feature with the maximum distance is selected as the optimal global feature. The extracted local features form a multivariate time series. To identify the most informative local features, a modified Mahalanobis distance [31] is employed for evaluation. The Mahalanobis

distances computed over the time series also form a new time series.

3) *Feature Enhancement*: Feature enhancement is performed on the original data using the optimal global and local features, resulting in a new sequence with improved discriminative capability. Specifically, the optimal global features are used to enhance the original data. The resulting enhanced feature sequence can be expressed as

$$\tilde{d}_G = d + \lambda(\alpha f_G - f_o)d \quad (3)$$

where  $f_G$  represents the optimal global feature values from normal data,  $f_o$  denotes the corresponding feature values from boundary samples,  $\alpha$  is the enhancement factor ( $\alpha > 1$ ), and  $\lambda$  is a tunable parameter that controls the enhancement strength.

Furthermore, feature enhancement is applied to the original data using the optimal local features. The corresponding enhanced sequence is denoted as

$$\tilde{d}_L = d + \lambda\alpha A^T (I + \lambda AA^T)^{-1} (f_L - Ad) \quad (4)$$

where  $f_L$  represents the optimal local feature values from normal data,  $A$  denotes the local feature extraction matrix,  $I$  is the identity matrix, and  $\alpha$  and  $\lambda$  retain their previous definitions.

### B. Reconstruction Model

Reconstruction-based anomaly detection methods require training a model that effectively captures the distribution of normal data. In this work, a VAE-LSTM architecture is adopted as the baseline model. A VAE is a generative model in which the encoder maps input data to a latent distribution, and the decoder reconstructs the original data from this representation. A well-known limitation of traditional AE is its tendency to overfit during training. To address this, the VAE encodes inputs as probability distributions—typically Gaussian—rather than deterministic points in the latent space. This stochastic formulation introduces beneficial randomness and serves as an implicit regularization mechanism, thereby enhancing model generalization. Given that satellite sensor data are time series in nature, employing LSTM as the underlying network enables the model to capture temporal dependencies and latent features more effectively. A two-layer LSTM architecture offers stronger nonlinear modeling capacity compared to a single-layer network. However, deeper architectures often yield diminishing performance gains while increasing computational overhead. Thus, a two-layer structure is adopted as a practical tradeoff.

VAE-LSTM is a fusion of VAE and LSTM. The VAE-LSTM model integrates the temporal modeling capability of LSTM with the probabilistic representation learning of VAE. This combination enables effective capture of complex temporal dependencies and latent distributions, making it particularly suitable for detecting subtle, evolving, or early-stage anomalies in time-series data. The encoder and the decoder of the VAE are constructed using LSTM networks. The deep model utilizes both the divergence loss and reconstruction loss to jointly

regularize the latent variable  $Z$ . The overall loss function of the deep model is defined as follows:

$$\text{Loss} = D_{\text{KL}}(q_\phi(z|x) \| p_\theta(z)) + \sum_{t=1}^N \text{mse}(x_t, \hat{x}_t). \quad (5)$$

The first term represents the Kullback–Leibler (KL) scatter between the latent variable distribution  $q(z|x)$  and the prior distribution  $p(z)$ , measuring how well  $q(z|x)$  represents the prior distribution  $p(z)$ . The second term computes the mean squared error between the model’s inputs and the original data series, representing the model’s reconstruction error.

The probability density function and KL divergence of a Gaussian distribution can be computed analytically, which greatly simplifies the optimization in variational inference. Therefore, the approximate posterior  $q(z|x)$  is typically assumed to follow a Gaussian distribution. When the complex distribution of  $x$  is mapped to the latent space through a neural network, the Gaussian assumption provides sufficient expressiveness to fit most application scenarios while significantly reducing computational cost.

We can map the input data  $x$  to a mean vector  $\mu$  and a variance vector  $\sigma$  via an encoder network, and then sample the latent variable  $z$  from the distributions of the mean and variance. Specifically, we can express it as follows:

$$\mu = f_\mu(x) \quad (6)$$

$$\sigma = f_\sigma(x) \quad (7)$$

$$z = \mu + \varepsilon \odot \exp(\sigma/2). \quad (8)$$

In the equation,  $\varepsilon \sim N(0, 1)$  represents a noise vector. To train the deep model, a backpropagation algorithm is employed. This algorithm iteratively updates the network parameters to minimize the defined loss function [32].

The Appendix presents a formal proof establishing the convergence of the VAE-LSTM reconstruction error for normal time-series data.

### C. Anomaly Detection Strategy

The globally enhanced data are reconstructed by the deep model to compute the reconstruction loss for each time point. The reconstruction loss quantifies the discrepancy between the original input and the reconstructed output at each time step. Reconstruction-based anomaly detection relies on this loss to determine whether an anomaly exists in the observed data. Two types of enhanced data—global and local—are passed through the reconstruction model, resulting in corresponding reconstruction losses. Accordingly, two different anomaly detection strategies are employed.

1) *Global-Scale Detection*: The formula for calculating the reconstruction loss is shown in (9). It is the arithmetic sum of the difference between the original data and the predicted data at each point

$$u = \sum_{i=1}^n (d(t_i) - \hat{d}(t_i)). \quad (9)$$

A larger sum of residuals reflects a greater overall discrepancy between the original data and the reconstructed data. Therefore, data with larger residual sums have a higher

probability of being anomalous. The maximum residual sum observed in normal data is used as the threshold for anomaly detection in that measurement.

2) *Local Feature Detection*: The global detection strategy is widely used but has inherent limitations. It can be viewed as projecting the entire time series at time  $t$  onto a single point, which inevitably results in some information loss. Therefore, we retain the entire residual sequence for each data point, as shown in the following equation:

$$D(t_i) = d(t_i) - \hat{d}(t_i). \quad (10)$$

The length of this sequence is the same as the test data, and if the whole residual sequence is tested directly, the redundant information in it will cause the results to be distorted. Therefore, we use 1-D average pooling to compress the residual sequence as in (10)

$$v = \frac{1}{h} \sum_{t=ks}^{ks+h} D(t) \quad (11)$$

where  $s$  represents the step size, while  $h$  signifies the width. Following the compression of the new data, we utilize support vector domain description (SVDD) as a means to differentiate between normal and abnormal data. The optimization objective of SVDD is to identify a minimum sphere characterized by a center  $\alpha$  and radius  $R$ , satisfying (12). Subsequently, we can utilize  $R$  as the threshold value for this detection measurement

$$(x_i - \alpha)^T (x_i - \alpha) \leq R^2 + \xi_i \quad \xi_i \geq 0. \quad (12)$$

3) *Anomaly Scores*: The anomaly scores derived from normal data by each detection method are transformed into anomaly scores within the range of  $[0, 1]$ . Among them,  $l_1$  denotes the anomaly score generated by global feature detection,  $l_2$  represents the anomaly score yielded by local feature detection, and  $l_3$  signifies the anomaly score originating from the original data acquired through the deep model. Their proportion of the anomaly score is determined by the characteristic scale factor  $\theta$ , whose value is calculated by the following equation:

$$\theta = \frac{1}{n} \sum_{t=1}^n \frac{\min_i \min_t |y(t) - x_i(t)| + \rho \max_i \max_t |y(t) - x_i(t)|}{|y(t) - x_i(t)| + \rho \max_i \max_t |y(t) - x_i(t)|} \rho = 0.5 \quad (13)$$

where  $y(t)$  represents the known boundary sample and  $x_i(t)$  is the data to be detected. The factor is calculated by the gray correlation analysis method and indicates the degree of similarity between the data to be detected and the boundary sample. The anomaly score is calculated as in (14). The detected data are recognized as anomalous when their anomaly score exceeds 1

$$L = \frac{(l_1 + l_2) l_3}{(1 - \theta)(l_1 + l_2) + 2\theta l_3} L \leq 1, \text{normal}. \quad (14)$$

The proposed framework consists of three independently structured deep submodels, each targeting a distinct feature perspective—global, local, and original. By applying a feature ratio-based weighting scheme to their outputs, the framework dynamically balances the contribution of each perspective and

reduces the risk of overfitting caused by the high concentration of boundary samples in any single feature space.

#### D. Model Update Mechanism

This study designs three model update strategies to enhance the adaptability and robustness of the model during long-term spacecraft operation. The first is a periodic update strategy, where the model is retrained at fixed time intervals using all newly collected data, allowing it to continuously adapt to changes in system status. The second is an event-triggered update strategy. When the system receives newly labeled boundary samples from domain experts, the model update is immediately triggered to enhance its capability in recognizing early anomalies. The third is an incremental update strategy. When the system detects concept drift in the data, only the global feature modeling component and the original data modeling component are incrementally updated, while the local feature detection component remains unchanged. This ensures local stability while improving the overall adaptability of the model.

In addition, this study proposes a method for the detection and early warning of concept drift. In practical deployment, telemetry parameters with limited variation ranges are first filtered out by engineers, allowing the system to focus on dynamic parameters that are more susceptible to drift. For these parameters, we continuously monitor the trend of anomaly scores. If a sustained deviation in the anomaly score is observed, concept drift is considered likely to have occurred, and the model update mechanism is subsequently triggered.

All the above update strategies enable timely correction of model parameters and dynamic adjustment of decision thresholds, thereby allowing adaptive tuning of the decision boundary and improving the models responsiveness to changes in system behavior.

#### E. Deployment

The proposed anomaly detection algorithm is primarily deployed at the ground control center to monitor and identify early anomalies during satellite operations. At fixed time intervals, the algorithm automatically retrieves specified key spaceborne sensor data from the ground-based data reception and processing system. Before being input into the model, the retrieved data undergo a series of preprocessing steps to improve data quality and enhance model robustness.

During the training phase, the anomaly detection framework supports the manual injection of boundary samples to strengthen the models ability to perceive early faults. Once trained, the model can automatically perform anomaly detection tasks, producing an anomaly label and a corresponding anomaly score for each data segment. When a potential anomaly is detected, the system immediately alerts operation and control personnel to assist in further analysis and intervention.

In addition, the framework also supports deployment on the satellite side to meet the real-time detection needs of certain critical sensor parameters. In such cases, the model can be trained on the ground and the updated parameters

uploaded to the spaceborne system via telemetry channels. The satellite-side system only performs lightweight inference, thereby balancing detection precision and system resource constraints and enhancing the satellites in-orbit autonomous anomaly recognition capability.

#### IV. EVALUATION OF THE EXPERIMENT AND ANALYSIS OF THE RESULTS

The experimental data were obtained from an in-orbit satellite. Over six months of continuous operation, voltage and current data from multiple channels were recorded by sensors monitoring the atomic clocks operational status. The data were sampled at 0.1 Hz. During in-orbit operation, several boundary samples that fell below the alarm threshold were identified by experts from the Chinese Space Agency. These samples serve as critical empirical evidence supporting this study.

The dataset contains 1894 time-series segments, among which 92 are labeled as anomalous samples. These annotations are based on engineering ground event logs and expert manual inspections, ensuring a high level of reliability. In the experimental setup, the first 300 normal segments are selected as the training set, and another 300 normal segments are used as the validation set. In addition, two rare boundary samples, identified by engineering experts, are included. The remaining data constitute the test set, which includes 1200 normal segments and the aforementioned 92 anomalous segments.

The anomalous samples cover three major types: point anomalies, contextual anomalies, and gradually evolving anomalies. These types collectively represent a wide range of abnormal behaviors that may occur during in-orbit satellite operations. Before being fed into the model, the raw spaceborne sensor data undergo several preprocessing steps, including data decoding, removal of transient outliers caused by single-event upsets, and interpolation for estimating missing values, in order to improve data continuity and usability.

##### A. Evaluation Metrics

The anomaly detection task in this study constitutes a typical imbalanced classification problem. Accordingly, the evaluation metrics used are recall, precision, and  $F1$ -score. Recall directly reflects the models ability to identify anomalous instances. In highly imbalanced datasets, precision more effectively reflects the impact of false positives on overall detection performance than the false positive rate (FPR). The  $F1$ -score provides a balanced measure of the models performance by combining precision and recall.

The definitions of these three metrics are given as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

$$F1 - \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (17)$$

##### B. Feature Extraction

We extracted 21 features for the training data. As can be seen in Fig. 3, the 13th feature has the highest discriminatory

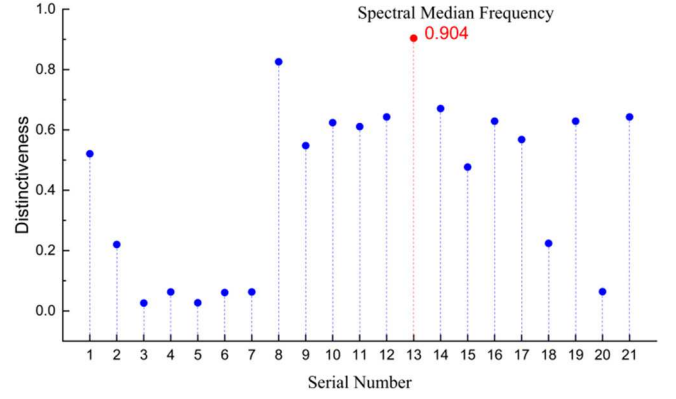


Fig. 3. Optimal global features.

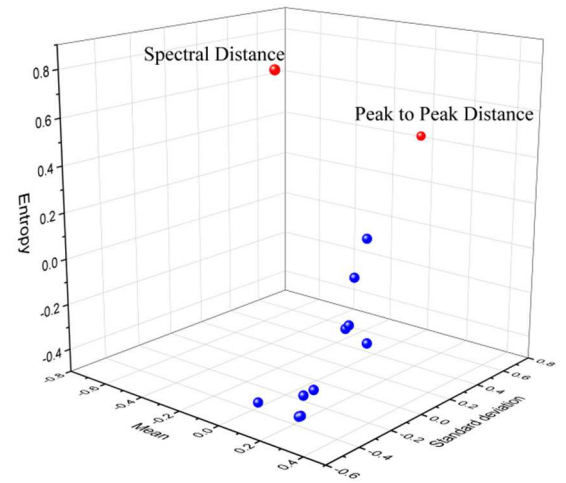


Fig. 4. Optimal local features.

degree, so this feature, i.e., the spectral median frequency (SMF) of the time series, is chosen as the best global feature.

For localized features, we use the Mahalanobis distance to measure the differentiation of different types of features. As can be seen in Fig. 4, two features, the spectral distance and the peak-to-peak distance, have greater differentiation as the best local features.

##### C. Comparative Experiment

To validate the effectiveness of the proposed MSFE-AD model, it is compared with four categories of anomaly detection approaches, including data enhancement, early anomaly detection, feature enhancement, and reconstruction-based approaches. A total of nine baseline methods, corresponding to representative studies in the literature, are selected for comparison.

The hyperparameters of all methods were individually tuned on the validation set using grid search. The hyperparameters for the proposed model were set as follows: number of LSTM units = 128, number of LSTM layers = 2, latent space dimension = 64, and KL divergence weight  $\beta = 1.0$ . To mitigate overfitting, dropout regularization was applied during training, along with an early stopping strategy based on the

TABLE II  
RESULTS OF PERFORMANCE COMPARISON ON  
IN-ORBIT TELEMETRY DATASETS

| Type                  | Method              | Recall       | Precision    | F1-score     |
|-----------------------|---------------------|--------------|--------------|--------------|
| Data                  | SMOTE               | 0.674        | 0.388        | 0.492        |
| Augmentation          | MAD-GAN             | 0.663        | 0.484        | 0.560        |
|                       | Transformer         | 0.696        | 0.604        | 0.647        |
| Early Detection       | parallel multiple   | 0.707        | 0.802        | 0.752        |
|                       | CNNs                | 0.696        | 0.587        | 0.637        |
| Feature Enhancement   | LSTM+Attention      | 0.696        | 0.587        | 0.637        |
|                       | FECAM               | 0.804        | 0.632        | 0.708        |
| Reconstruction Method | VAE                 | 0.685        | 0.649        | 0.667        |
|                       | LSTM-AE             | 0.761        | 0.909        | 0.828        |
|                       | LSTM-VAE            | 0.772        | 0.877        | 0.821        |
|                       | The proposed method | <b>0.826</b> | <b>0.938</b> | <b>0.878</b> |

TABLE III  
RESULTS OF ABLATION EXPERIMENTS EVALUATION INDICATORS

| Method                            | Recall       | Precision    | F1-score     |
|-----------------------------------|--------------|--------------|--------------|
| Global Feature Enhancement Method | 0.783        | 0.706        | 0.743        |
| Local Feature Enhancement Method  | 0.772        | 0.772        | 0.772        |
| Feature Enhanced LSTM-VAE         | <b>0.837</b> | 0.694        | 0.759        |
| The Proposed Method               | 0.826        | <b>0.938</b> | <b>0.878</b> |

validation loss. This enhances the models generalization ability. The final configuration was selected by balancing detection precision and computational efficiency. In addition, NumPy, scikit-learn, and Pandas libraries are used. The final result of each anomaly detection algorithm is the average score of ten repeated runs. Table II shows the recall, precision, and F1-score results of the ten experiments on real satellite data.

#### D. Ablation Studies

To further analyze the effectiveness of the proposed model, ablation experiments are conducted. Different combinations of model components are evaluated to observe their impact on anomaly detection performance. Each configuration is integrated with the VAE-LSTM deep reconstruction model.

First, global feature enhancement is applied in conjunction with global-scale anomaly detection. Second, local feature enhancement is combined with local-scale detection. Third, both combinations are employed simultaneously; if either strategy detects an anomaly, the data are classified as anomalous.

The corresponding results are presented in Table III.

Our observations indicate that, compared to the baseline model, all three methods improve recall. The third combination achieves comparable recall performance to the MSFE-AD model. However, its precision and F1-score are substantially lower than those of the MSFE-AD model. This result is illustrated in Fig. 5.

These findings demonstrate that the use of feature enhancement methods along with their corresponding detection strategies can effectively enhance the detectors capability to identify anomalous data. The anomaly score evaluation mechanism helps eliminate false positives and improves the models robustness. This validates that each component of our approach is indispensable.

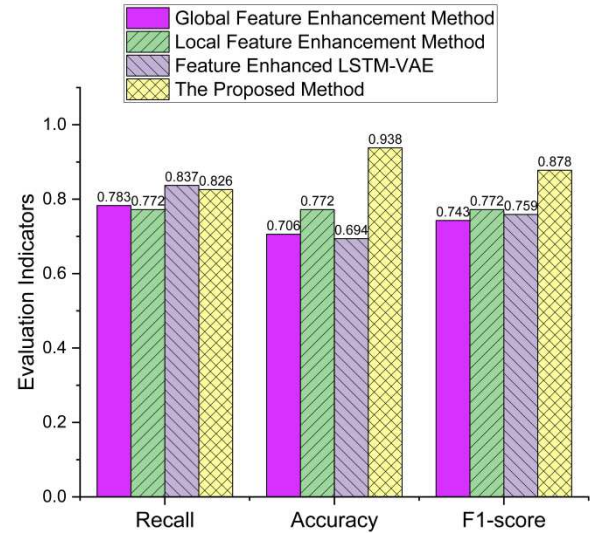


Fig. 5. Comparison of the feature-enhanced LSTM-VAE with the proposed method on three key evaluation metrics.

TABLE IV  
AVERAGE RESULTS OF EACH TYPE OF METHOD

| Type                  | Recall       | Precision    | F1-score     |
|-----------------------|--------------|--------------|--------------|
| Data Augmentation     | 0.669        | 0.436        | 0.526        |
| Early Detection       | 0.702        | 0.703        | 0.700        |
| Feature Enhancement   | 0.750        | 0.610        | 0.673        |
| Reconstruction Method | 0.739        | 0.812        | 0.772        |
| Proposed methodology  | <b>0.826</b> | <b>0.938</b> | <b>0.878</b> |

## V. DISCUSSION

### A. Analysis and Exploration

In the experimental results, we observe that some baseline methods exhibited lower precision. This is because there is a significant class imbalance, with many more normal samples than abnormal ones. Consequently, when some normal samples are misclassified as anomalies, precision deteriorates. Nine anomaly detection methods, categorized into four types, were used for comparison. Table IV presents the average results for each category. Results indicate that feature enhancement and reconstruction-based methods achieved comparatively superior performance. Building upon this, the methods proposed in this article further improve anomaly detection capability. This result is illustrated in Fig. 6.

The ablation experiments demonstrate that feature enhancement significantly improves the models ability to detect anomalous data, as evidenced by an increased recall. This indicates that the proposed feature enhancement method utilizing boundary samples is more sensitive to anomalies. However, this comes at the cost of increased false positives, leading to reduced precision and F1-score. To address this issue, a learnable scaling factor is introduced within the anomaly detection framework to balance the contributions of different components. While this adjustment slightly reduced recall, it yielded the best precision and F1-score.

These results clearly demonstrate that the application of appropriate feature engineering techniques and anomaly

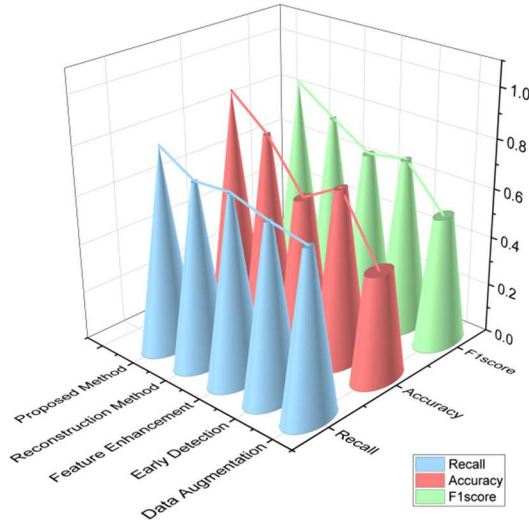


Fig. 6. Average results of each type of method.

detection strategies, including the anomaly score evaluation mechanism, has a substantial impact on detection performance.

### B. Stability Analysis of the Multiscale Detection Score Fusion Method

To ensure the reliability of the multiscale detection score fusion results, it is necessary to analyze the stability of the fusion (14). On the one hand, such analysis prevents the output from diverging or exhibiting abnormal fluctuations when the inputs remain within a reasonable range, thereby avoiding misjudgments in anomaly detection. On the other hand, demonstrating that the output variations remain bounded under small input perturbations improves the overall robustness of the system, ensuring that the proposed fusion method possesses practical applicability in real-world scenarios.

First, the input to the fusion formula is a bounded set composed of detection scores from different time scales. Accordingly, the input parameters are bounded. Since the denominator  $D(\theta)$  remains strictly positive within the interval  $\theta \in [0, 1]$ , inequality-based derivations confirm that the output  $L(\theta)$  is also bounded. This demonstrates that the fusion function satisfies the bounded-input bounded-output (BIBO) stability condition, implying that the fusion output does not diverge when the input scores are confined to a finite range, thereby preventing abnormal fluctuations during the decision-making process.

Furthermore, in terms of system convergence, if the detection scores of the subdetectors converge, the limit of the fusion output  $L$  necessarily exists due to the commutativity of limits for continuous functions. This establishes that the overall system exhibits asymptotic stability. From the perspective of local robustness, algebraic simplification and upper bound estimation of the partial derivatives of the fusion function indicate that the function satisfies Lipschitz continuity on a specific compact set. Consequently, small perturbations in the input (e.g., those caused by significant variance across time scales) lead to bounded variations in the output—this

further ensures the local stability of the fusion process and, thus, avoids erratic fluctuations in the final anomaly score and decision boundaries.

### C. Selection and Evaluation of Deep Models

The proposed anomaly detection framework adopts a modular design, in which key components such as feature augmentation, time-series modeling, and anomaly scoring are decoupled. This allows flexible replacement and optimization of submodules while maintaining the overall structural stability. Currently, time-series modeling is implemented using a VAE-LSTM model. In extensive comparative experiments, this model demonstrated outstanding performance in several key metrics, including precision, recall, and F1-score, and was, therefore, selected as the default modeling component in the framework.

However, we also acknowledge that model selection based solely on performance metrics typically relies on known anomalous samples, which introduces certain limitations. Specifically, a “best-performing model” selected based on existing anomaly types may perform poorly when encountering previously unseen anomalies. This evaluation strategy is essentially conducted under the closed-world assumption, which may fail to fully reflect the model’s robustness and generalization ability in real-world open environments.

To address this issue, we plan to explore more general-purpose model evaluation mechanisms in the future, with the aim of enhancing the reliability and practical guidance value of the model in real-world applications.

### D. Interpretability Analysis

The interpretability of deep learning methods has long been a major challenge in practical applications. In the proposed framework, feature engineering is applied before training the deep learning model to extract a series of features with clear physical meanings from the raw telemetry data. These features capture the variation patterns and statistical properties of telemetry signals across different dimensions. This domain knowledge-driven feature construction enhances the interpretability of detection results, making the model outputs easier for experts to understand and verify.

### E. Computational Efficiency and Deployment Feasibility

Although the proposed model achieves promising performance in anomaly detection tasks, its computational efficiency is also an important consideration. On a computer configured with an Intel i5-12400 CPU, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 Ti GPU, the model takes approximately 85 s to complete inference on the entire dataset. Compared with LSTM-based models, the inference time is slightly longer; however, it demonstrates a significant advantage in computational efficiency over models based on GAN or transformer architectures. This advantage becomes more pronounced as the input sequence length increases.

Similar to other deep learning-based anomaly detection methods, the computational cost of this model during training

is substantially higher than during inference. Fortunately, ground-based management systems typically have sufficient computing resources to support the training of complex models. In contrast, due to limited resources on the satellite side, only the inference model and necessary detection modules are usually deployed. With the help of model compression techniques, such as quantization and pruning, the computational and storage cost of deploying the model in spaceborne systems can be further reduced. In addition, for models with parallel structures, a rotational working mechanism can be adopted to reduce continuous computational load, at the expense of some real-time performance.

#### F. Applicability and Limitations

The proposed method focuses on detecting early faults in satellites, which is of great significance for enhancing mission safety and extending satellite service life. The framework adopts a multiscale parallel heterogeneous model along with a corresponding anomaly scoring scheme, aiming to ensure applicability beyond a single scenario. Our experimental dataset includes point anomalies, contextual anomalies, and gradually evolving anomalies. The results demonstrate that the proposed method exhibits strong competitiveness under the current experimental setting.

However, for more complex anomaly patterns (e.g., semantic anomalies), the discriminative ability of the existing framework may be limited in the absence of sufficient domain knowledge. For certain satellites with high maneuverability—such as military reconnaissance satellites or technology demonstration satellites—their sensor data may exhibit significant nonstationarity and complex variations due to frequent trajectory changes and dynamic task adjustments. In such cases, the framework may face performance constraints.

In addition, mislabeled anomalies in the training data, insufficient training samples, or a lack of awareness of abrupt changes in mission conditions may result in biased modeling and degraded detection performance.

## VI. CONCLUSION

During spacecraft operation and maintenance, a very small number of boundary samples labeled manually by domain experts exist. These samples cannot be detected by existing anomaly detection methods. Such boundary samples lie near the classification decision boundary and are highly similar to both normal and anomalous data. This article proposes an anomaly detection framework that constructs a highly discriminative feature space using boundary samples. The method significantly enhances the separability of normal and anomalous samples in the feature distribution, thereby effectively reducing the learning difficulty of neural networks in early anomaly detection tasks and improving classification precision.

By introducing feature weighting factors to dynamically fuse the outputs of multiscale parallel heterogeneous sub-models, the framework achieves a dynamic balance across different feature perspectives. This effectively mitigates the

overfitting risk caused by clustering of boundary samples. On a real satellite sensor dataset, the proposed method achieves excellent detection performance: a recall rate of 0.826, reflecting strong recognition of anomalous events; a precision of 0.938, indicating overall reliable judgment; and an  $F1$ -score of 0.878, demonstrating a balanced tradeoff between precision and recall. Compared with other state-of-the-art methods, the proposed anomaly detection framework attains the best results.

The proposed anomaly detection model addresses specific engineering application scenarios. The validation data were collected from spacecraft payloads experiencing gradual degradation. Such data collection is challenging and requires manual annotation by domain experts. More data are needed in the future to further validate the model. The proposed method has a clear underlying principle. As a general framework, its deep learning model component can be replaced. Future work will focus on improving evaluation mechanisms, exploring more robust and generalizable deep models, and further strengthening the collaborative design between deep models and anomaly detection strategies to enhance overall detection performance and application reliability.

## APPENDIX

In order to validate the convergence of the reconstruction error of the VAE-LSTM model on normal time-series data, the derivation is divided into the following three key steps.

### A. Step 1: Proof of Monotonic Nonincreasing Property Based on the Nonnegativity of KL Divergence

The total loss of VAE-LSTM is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \beta \cdot \text{KL}(q||p) \quad (18)$$

where  $\beta = 1$ , and  $q$  and  $p$  denote the posterior and prior distributions of the latent variable  $z$ . According to the fundamental property of KL divergence, namely, for any probability distributions  $q$  and  $p$ ,  $\text{KL}(q||p) \geq 0$ , it follows that  $\mathcal{L}_{\text{rec}} \leq \mathcal{L}_{\text{total}}$ .

During training, the Adam optimizer minimizes the total loss via gradient descent. Thus, for any adjacent iterations  $n$  and  $n + 1$ , it holds that  $\mathcal{L}_{\text{total}}(n + 1) \leq \mathcal{L}_{\text{total}}(n)$ . Combining the inequalities, we obtain

$$\mathcal{L}_{\text{rec}}(n + 1) \leq \mathcal{L}_{\text{total}}(n + 1) \leq \mathcal{L}_{\text{total}}(n) \quad (19)$$

$$\mathcal{L}_{\text{rec}}(n) \leq \mathcal{L}_{\text{total}}(n). \quad (20)$$

The regularization term is nondecreasing during training; therefore,  $\mathcal{L}_{\text{rec}}(n + 1) \leq \mathcal{L}_{\text{rec}}(n)$ , which demonstrates that the reconstruction error exhibits a monotonic nonincreasing property.

### B. Step 2: Proof of Lower Bound Existence Based on the ELBO of VAE

For normal time-series data  $x$ , its log-likelihood  $\log p(x)$  is constant since the distribution of normal data is fixed. It satisfies the evidence lower bound (ELBO) inequality

$$\log p(x) \geq E_q[\log p(x|z)] - \text{KL}(q||p) \quad (21)$$

where  $E_q[\log p(x|z)]$  denotes the expected log-likelihood of  $x$  generated from the latent variable  $z$ , which is negatively

correlated with the reconstruction error  $\mathcal{L}_{\text{rec}}$ . Rearranging the inequality yields

$$\mathcal{L}_{\text{rec}} \geq -(\log p(x) + \text{KL}^*) \quad (22)$$

where  $\text{KL}^*$  is the stabilized value of the KL divergence at the later stage of training. Let

$$C = -(\log p(x) + \text{KL}^*) > 0 \quad (23)$$

constrained by inherent noise in time series. Therefore,  $\mathcal{L}_{\text{rec}} \geq C$ , indicating that the reconstruction error has a valid lower bound.

### C. Step 3: Proof of Asymptotic Stability Based on Gradient Descent Convergence

The derivation is based on the following two assumptions.

- 1) The total loss  $\mathcal{L}_{\text{total}}$  satisfies Lipschitz continuity—this is ensured by the bounded spectral norm of LSTM weight matrices and the Lipschitz property of the ReLU activation function.
- 2) The gating mechanism of LSTM effectively alleviates the vanishing gradient problem, ensuring that gradients can be propagated back through time-series data.

The learning rate  $\eta_n$  of the Adam optimizer satisfies the Robbins–Monro convergence conditions:  $\sum \eta_n = \infty$  (divergent sum of step sizes) and  $\sum \eta_n^2 < \infty$  (convergent sum of squared step sizes). Hence, the total loss converges to a local minimum, i.e.,

$$\lim_{n \rightarrow \infty} \mathcal{L}_{\text{total}}(n) = \mathcal{L}_{\text{total}}^* \quad (24)$$

Combining the monotonic nonincreasing property and the existence of a lower bound of the reconstruction error, the monotone convergence theorem further implies that

$$\lim_{n \rightarrow \infty} \mathcal{L}_{\text{rec}}(n) = \mathcal{L}_{\text{rec}}^* \quad (25)$$

where  $\mathcal{L}_{\text{rec}}^*$  denotes the stable limit of the reconstruction error, which proves the asymptotic stability of reconstruction error convergence.

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